Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells

Jonathan Mace*

Microsoft Research, Redmond, WA, United States

Brian R. Mace

Acoustics Research Centre, Department of Mechanical and Mechatronics Engineering, University of Auckland, Auckland, New Zealand

Abstract

The resonant behaviour of fluid-filled viscoelastic shells arises repeatedly in biomechanics, postharvest sensing, and non-destructive evaluation, yet each application is typically treated in isolation. This paper advances a unifying thesis: *shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem.*

We formalise this thesis through four formal results and a supporting excitation proposition. First, we prove that equivalent-radius (spherical) reductions of non-spherical shells are *rank-deficient* in the sense of structural identifiability: geometry and material parameters become confounded, rendering independent recovery impossible. Second, we establish that oblate asphericity lifts this rank deficiency through curvature-dependent mode splitting, providing the additional spectral information needed for locally identifiable inversion. Third, we demonstrate the corrected near-spherical asymptotics, $\sigma_3(\varepsilon) = \lambda_1\varepsilon^2 + O(\varepsilon^4)$, so the exact sphere is singular and the previously reported finite floor is recognised as a discretisation artefact. Fourth, we construct a counterexample showing that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere model may predict resonant frequencies within engineering tolerance yet fail severely when used for parameter estimation.

These formal results are supported by cross-application evidence drawn from the human abdomen (infrasound resonance), the watermelon (acoustic ripeness assessment), and the urinary bladder (fill-dependent resonance), demonstrating that geometry is the hidden variable unifying apparently disparate vibroacoustic phenomena.

Keywords: fluid-filled shells; viscoelastic resonance; structural identifiability; inverse problems; asphericity; Kac problem

*Corresponding author. E-mail: jonathanmace@microsoft.com

1 Introduction

Spectral inverse problems ask what physical parameters can be recovered from an observed resonance spectrum. Versions of that question arise in structural dynamics, elastographic parameter estimation, and non-destructive testing, where one compares measured modal data with a forward model and seeks the parameter set that best explains the observations [4, 5, 23, 3, 2]. The central issue is not simply whether a model reproduces the spectrum in the forward direction. It is whether the spectrum contains enough independent information to distinguish the geometric and material parameters that generated it. Following the classical identifiability framework of Bellman and Åström [1], Rothenberg [26], and its computational operationalisation by Raue *et al.* [25], the question is one of both rank and conditioning.

For fluid-filled viscoelastic shells, geometry enters that question in three distinct roles. It filters how external forcing projects onto admissible deformation fields; it organises the modal spectrum through curvature, symmetry, and fluid loading; and it determines whether that spectrum supports identifiable parameter recovery. The first role explains why a resonance can exist yet remain weakly excitable. The second explains why different modal families respond differently to shape. The third is the main concern of the present paper, because it determines whether spectral data can separate geometric and material effects at all. Figure 1 summarises these three geometric roles and their relation to the present shell models.

The analogy with Kac's question is instructive but limited. Kac's 1966 formulation remains a canonical statement of the inverse spectral ambition [8]. The present problem concerns local parametric identifiability of a finite-dimensional model, not global spectral uniqueness of a differential operator. The operator is a coupled fluid–structure system rather than a scalar Laplacian, but the analogous issue remains which geometric features are spectrally distinguishable. The Gordon–Webb–Wolpert counterexample remains relevant as a reminder that symmetry can hide geometric information even when spectral data are exact [6].

We study these issues for fluid-filled viscoelastic shells. Such shells arise in biomechanics, soft-structure acoustics, postharvest sensing, and related inverse problems, yet their spectral interpretation is often application-led. Here structural identifiability refers to local parameter distinguishability encoded by Jacobian rank, whereas practical identifiability refers to the conditioning of that recovery problem under finite perturbations and finite-precision data [25, 2]. The scaled Jacobian framework used in Section 3 belongs directly to that tradition.

Within that framework, the most restrictive mechanism is geometric reduction. Symmetry-reduced descriptions, especially equivalent-radius reductions, can be entirely serviceable as forward surrogates because they preserve the leading frequency scale of a non-spherical shell. Their inverse behaviour is less forgiving. Once two geometric degrees of freedom are compressed into a single scalar descriptor, the spectrum may lose an independent curvature channel and the scaled Jacobian may become rank deficient. Structural identifiability is then lost. Even when rank is formally restored, the smallest singular value may remain small enough that the inverse problem is severely ill-conditioned.

Oblate asphericity can restore local rank in the present setting, but only in the precise local sense established in Section 3. Proposition 3 does not assert that arbitrary symmetry breaking is sufficient. It shows, more modestly and more usefully, that oblate asphericity can lift the rank deficiency on a non-empty open set because different flexural modes sample meridional and circumferential curvature differently. Proposition 4 then gives the corrected near-spherical law

$\sigma_3(\varepsilon) = \lambda_1\varepsilon^2 + O(\varepsilon^4)$, so the exact sphere is singular and local distinguishability grows regularly as the shell departs from spherical symmetry. At the canonical oblate operating point, the inverse is not merely full-rank but moderately conditioned, with $\kappa = 69.4$ for the five-mode study reported in [16].

The applications considered here provide evidence for that broader claim. They include human abdominal resonance, beginning from an infrasound-coupling question in abdominal biomechanics (including the phenomenon sometimes discussed under the "brown note" label) [14, 21]; watermelon ripeness assessment, where a reduced forward surrogate can remain useful when the inferential target is deliberately narrow [15]; and urinary-bladder resonance, where fill state changes both size and effective stiffness and therefore provides a contrasting geometry-sensitive shell application [17]. The formal results stated below are new; the applications are drawn from companion studies and serve as the evidence base.

The formal development contains four principal formal results and one supporting excitation proposition. The supporting proposition, Proposition 1, recasts the Breit–Wigner excitation filter as a geometric statement about modal coupling. The four principal results are these. Theorem 2 proves rank collapse under scalar geometric reduction and therefore the loss of structural identifiability. Proposition 3 shows that oblate asphericity can restore full rank on a non-empty open set. Proposition 4 gives the correct near-spherical asymptotics for the smallest singular value of the scaled Jacobian. Proposition 6 then demonstrates that small forward spectral error does not by itself imply an inverse model suitable for parameter recovery.

The broader significance is not confined to the present shell class. Whenever resonance data are used to infer geometry, stiffness, or both, there is an unwarranted inference to avoid: success in the forward fit does not by itself establish that the inverse map is informative. The identifiability lesson is that symmetry can preserve the dominant spectral scale while removing the independent geometric channels required for parameter recovery.

Our scope is deliberately local and linearised. We study structural identifiability through Jacobian rank and practical identifiability through the singular values and condition number of a scaled Jacobian built from modal frequencies. We do not claim global uniqueness for nonlinear inverse problems, and we do not suggest that spectral data alone resolve every ambiguity in real shells. The analysis is restricted to linearised shell dynamics with homogeneous viscoelastic walls, axisymmetric reference geometries, and a single internal fluid phase; experimental routes are discussed later, but no new experiments are presented here.

Section 2 recalls the companion evidence and the common geometric structure required for the present argument. Section 3 proves the supporting proposition and the four principal formal results within a unified local-identifiability framework. Sections 4 and 5 test and interpret those results across the application record, and Section 6 closes with the general lesson for spectral inference on shells.

2 Background

2.1 Shell-model overview

The companion papers share a common forward model: a fluid-filled viscoelastic shell whose undeformed midsurface is represented, to leading order, as an oblate spheroid with semi-major axis

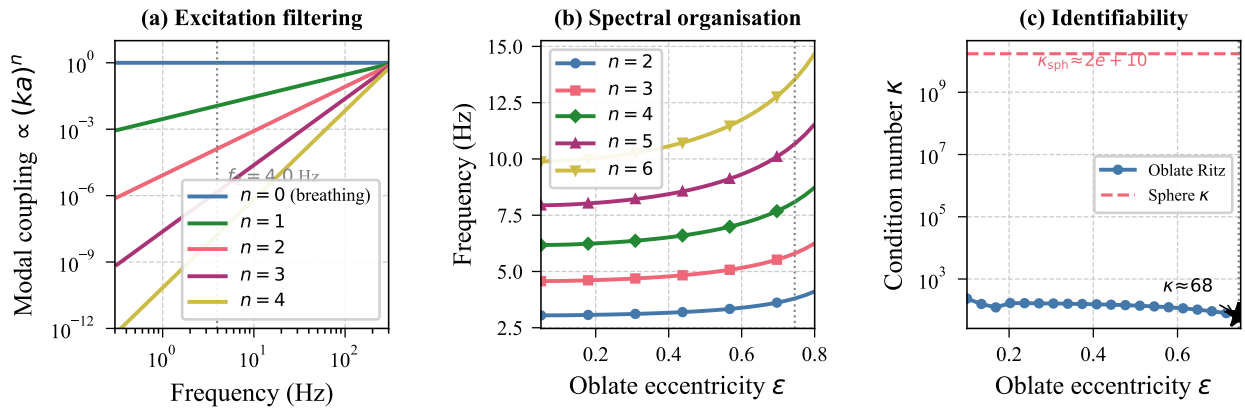

Figure 1: The three roles of geometry in fluid-filled shell mechanics: (a) excitation filtering via $(ka)^{2n}$ suppression, (b) spectral organisation showing modal frequency dependence on eccentricity, (c) identifiability improvement showing condition number reduction from sphere to oblate Ritz model.

a and semi-minor axis c. The modelling strategy follows the classical shell-acoustics tradition from Lamb's treatment of elastic-shell modal families [10], through Love–Kirchhoff shell theory [12], to later fluid–structure formulations in which structural inertia, fluid loading, and curvature appear on equal footing [7]. The abdominal formulation in [14] adapted that framework and used a Rayleigh–Ritz approximation to retain the features that matter for the present programme: curvature, wall compliance, internal loading, and mode shape. A self-contained summary of the Ritz basis, boundary conditions, and convergence evidence is provided in Appendix A. The capstone uses the resulting structure as the common language in which the subsequent applications can be compared.

In schematic form, if $\mathbf{q}$ denotes the vector of Ritz amplitudes, the kinetic and potential energies are written as

$$T = \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\left(\mathbf{M}_s + \mathbf{M}_f\right)\dot{\mathbf{q}}, \qquad V = \frac{1}{2}\mathbf{q}^{\mathsf{T}}\mathbf{K}\mathbf{q}, \tag{1}$$

where $\mathbf{M}_s$ is the shell inertia matrix, $\mathbf{M}_f$ is the added-mass contribution from the entrained fluid, and $\mathbf{K}$ is the effective stiffness matrix arising from membrane strain, bending, and any prestress retained in the model. Stationarity of the Rayleigh quotient then leads to the generalised eigenvalue problem

$$\left[\mathbf{K} - \omega^2\left(\mathbf{M}_s + \mathbf{M}_f\right)\right]\mathbf{q} = 0, \tag{2}$$

where ω is the angular natural frequency. For nearly spherical shells this structure reduces to the classical modal families identified by Lamb [10]; for the more general fluid-loaded case, it fits squarely within the coupled shell–acoustic framework described by Junger and Feit [7]. The important point for the present paper is that geometry enters twice. It shapes the admissible deformation fields through the basis itself, and it enters the energetic balance through curvature-dependent stiffness and fluid coupling. The same geometric ingredients later control both forcing efficiency and inverse conditioning.

2.2 Companion studies: Stage I — the question

Stage I asked whether the brown-note claim contained any physically defensible content, and the resulting analysis proved more informative than the original claim itself. The abdominal model in [14], formulated as a fluid-filled oblate shell, possesses a genuine low-frequency flexural resonance, with the second mode occurring at

$$f_2 = 3.95\,\mathrm{Hz}. \tag{3}$$

That result mattered because it separated two questions that are often conflated: whether a resonance exists, and whether any realistic forcing path can excite it appreciably. The modal calculation answered the first question in the affirmative. The coupling analysis answered the second in the negative for airborne infrasound. Because the acoustic field is both long-wavelength and weakly matched to the relevant flexural deformation, the airborne pathway does little work on the shell. Mechanical forcing through whole-body vibration projects onto the same mode family far more effectively, producing a coupling ratio of approximately $R \approx 3.3 \times 10^4$ between the mechanical and airborne responses [14]. The resonance was genuine, but the direct acoustic pathway was ineffective.

The first important refinement, reported in [18], considered tissue-constrained gas pockets as local compliant inclusions capable of much stronger pressure-to-motion transduction. This did not contradict the whole-cavity result of [14]. Rather, it refined the interpretation. If one seeks a global flexural response of the abdominal shell, airborne excitation is ineffective. If one asks whether smaller substructures embedded within the abdomen might respond locally, gas pockets are more plausible candidates. The programme therefore began with a division that remains useful throughout the capstone: global shell resonance organises the large-scale spectrum, but local resonant substructures may dominate transduction when coupling efficiency is the relevant physiological question.

2.3 Stage II — evidence across scales, organs, and pathways

Stage II tested whether the same shell-theoretic logic survived changes in scale, organ, and excitation pathway. The dimensionless reformulation in [20] recast the abdominal resonance problem and derived scaling laws that made the low-order flexural behaviour comparable across species [20]. Its point was not merely zoological. The governing frequency scales can be expressed through a compact set of geometric and mechanical groups, which shows that the infrasonic shell problem is structured by similarity, rather than by the contingencies of one anatomical example.

The urinary-bladder analysis in [17] transferred the framework and thereby tested it on an organ whose geometry changes substantially with physiological state [17]. Fill-dependent resonance in the bladder cannot be interpreted as a simple size effect. Increasing volume changes characteristic length scales, but it also alters wall tension and effective stiffness. The resulting modal trajectory therefore reflects a competition between geometry and stiffening, not a monotonic geometric trend. This was the first clear sign, outside the original abdominal problem, that a single scalar reduction of shape is unlikely to preserve the physically relevant sensitivities.

The borborygmi study in [13] shifted attention from infrasonic flexure to an audible internal phenomenon. Its constrained-bubble model produced characteristic frequencies in the range 135 Hz to 440 Hz, matching the clinical band of gut sounds. The mechanism was distinct from

whole-cavity flexure, but the shell still mattered. It set the confinement and loading conditions within which local gas pockets oscillate. The paper therefore broadened the programme without abandoning its central theme: geometry continues to mediate what kind of spectrum is possible, even when the relevant source is local rather than global.

The sub-bass analysis in [21] returned to low frequencies and extended the coupling question to the sub-bass regime, where concert exposures and whole-body vibration are often discussed together despite acting through rather different pathways [21]. The conclusion was consistent with the abdominal analysis in [14]. Airborne sub-bass remains a poor driver of substantial abdominal motion, whereas structure-borne excitation can couple much more efficiently. By the end of Stage II, the programme had assembled a coherent evidential base: across species, organs, audible and infrasonic ranges, and airborne and mechanical forcing, geometry continued to organise the modal spectrum while filtering how energy enters it.

2.4 Stage III — from forward adequacy to inverse structure

Stage III turned from forward resonance prediction to the inverse question of what one can infer from a measured spectrum. Watermelon tap-tone assessment in [15] provided the constructive example. An equivalent-sphere shell model can be forward-adequate for a practical inversion task when the inferential target is narrow and the geometric uncertainty is limited. In that context the model need only map a measured resonance to an effective rind stiffness with useful fidelity. These results therefore show that reduced geometry can be serviceable when the application asks for classification or approximate property recovery, rather than full geometric disentanglement.

The Kac-style identifiability analysis in [16] exposed the inverse cost of that simplification. Asked in Kac-like language whether one can "hear" the shape of an organ, the programme's answer became conditional rather than categorical. If the shell is reduced to a single equivalent radius, material and geometric sensitivities become strongly correlated, and the inverse map is nearly singular. The condition-number analysis established that the equivalent-sphere model is severely ill-conditioned, whereas the full oblate-shell model restores practical separation of parameter effects. Just as importantly, [16] revealed an oblate–prolate asymmetry: breaking spherical symmetry is not, by itself, enough. What matters is whether the perturbation changes how different modes sample curvature.

A short theorem-oriented note on identifiability lifting [19] condensed that observation. Its contribution was to show that oblate deformations lift the near-spherical inverse degeneracy in a way that prolate deformations do not improve comparably. This matters to the capstone because it rules out an imprecise summary. One cannot simply say that "asphericity helps". Some geometric departures from the sphere decorrelate the spectral sensitivities that matter for inversion; others do not. The programme's theoretical turn therefore culminated in a sharper claim: forward adequacy and inverse adequacy are distinct, and their separation is governed by geometry.

2.5 Broader context

The Browntone series sits at the meeting point of several established literatures. One is classical shell acoustics, in which curvature, modal families, and fluid loading have long been treated as inseparable elements of the forward problem [12, 11, 28, 7]. Another is spectral inverse theory, from Kac's celebrated question and the Gordon–Webb–Wolpert counterexample to engineering

treatments of vibration inversion and model updating [8, 6, 4, 5, 23, 3]. A third is the practical language of identifiability, familiar from inverse elasticity and biomedical imaging, in which the issue is not merely uniqueness in principle but whether the measured observables contain enough independent information to distinguish the parameters of interest under realistic sensitivity and noise structures [25, 2, 24, 27].

What the present programme adds is a concrete family of case studies in which these literatures can be read together. The same geometric framework explains why a soft organ can possess a low-frequency shell resonance yet remain almost immune to direct airborne excitation; why local gas pockets and gut sounds are better interpreted as shell-constrained substructures than as whole-cavity breathing modes; why a reduced model can succeed as a forward surrogate for fruit-ripeness assessment; and why that same reduction can fail badly when used for multi-parameter inversion. In biomedical and engineering settings alike, the practical question is rarely whether resonance exists in the abstract. It is whether the observable spectrum contains geometry-specific information that survives modelling reduction and experimental uncertainty. The companion papers suggest that this question is best answered not by separating shell mechanics, acoustics, and inversion into different problems, but by treating geometry as the common mediator of all three.

3 Theory

The prior work developed thus far supplies three ingredients that are required in the present section. First, the abdominal analysis separates the low-frequency flexural family from the much higher breathing mode in a fluid-filled soft shell and identifies geometry as the quantity that controls not only resonance but also the efficiency with which forcing projects onto those modes [14]. Secondly, the watermelon inversion provides a case in which an equivalent-sphere model remains useful because the inferential target is deliberately narrow and the geometry is known independently [15]. Thirdly, the identifiability analysis shows the inverse cost of that reduction: once geometry and material must be inferred simultaneously, the same equivalent-radius simplification becomes structurally defective [16]. The present section turns those ingredients into formal statements: one proposition for excitation filtering and four proved results for the inverse problem.

We emphasise at the outset that the statements below concern *local practical identifiability* of flexural spectra. They do not claim global uniqueness for arbitrary nonlinear inversions. They ask a more disciplined question: given a nominal operating point, do the measured modal frequencies contain enough linearly independent information to distinguish changes in geometry from changes in material stiffness?

3.1 Common setup

Consider a fluid-filled viscoelastic shell with semi-major axis a, semi-minor axis c, Young's modulus E, and fixed secondary parameters

$$\mathbf{p} = (h, \nu, \rho_w, \rho_f, K_f, P_{\mathrm{iap}}, \eta).$$

For the Browntone applications, the canonical values are $E = 0.1\,\mathrm{MPa}$, $a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 0.01\,\mathrm{m}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$, $K_f = 2.2\,\mathrm{GPa}$, $P_{\mathrm{iap}} = 1000\,\mathrm{Pa}$, and

$\eta = 0.25$ [14, 16]. We restrict attention to the flexural modes $n \geq 2$, since the breathing mode $n = 0$ is fluid-compressibility dominated and lies in the kilohertz range for the canonical abdomen rather than the single-digit-hertz range of interest [14]. That distinction is essential for physical consistency.

Let the unknown parameter vector be

$$\boldsymbol{\theta} = (a, c, E)^\top \in \Theta \subset \mathbb{R}^3_+, \tag{4}$$

and let the measured modal set be $\mathcal{M} = \{n_1, \ldots, n_m\}$, with $m \geq 3$. The forward map is

$$\mathbf{f}(\boldsymbol{\theta}) = \big(f_{n_1}(\boldsymbol{\theta}; \mathbf{p}), \ldots, f_{n_m}(\boldsymbol{\theta}; \mathbf{p})\big)^\top \in \mathbb{R}^m, \tag{5}$$

where f_n is the natural frequency of flexural mode n. The raw Jacobian is

$$J_{ij} = \frac{\partial f_{n_i}}{\partial \theta_j}, \tag{6}$$

but because the three parameter directions carry different units, the relevant local sensitivity operator is the scaled Jacobian

$$\widetilde{J}[i, j] = \frac{\partial f_{n_i}}{\partial \theta_j} \frac{\theta_j}{f_{n_i}}. \tag{7}$$

Each entry of $\widetilde{J}$ is dimensionless and measures fractional frequency change per fractional parameter change. Let

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0 \tag{8}$$

denote the singular values of $\widetilde{J}$, and define the local conditioning metric

$$\kappa(\widetilde{J}) = \frac{\sigma_1}{\sigma_3}. \tag{9}$$

When $\sigma_3 = 0$, the inverse problem is locally rank deficient. When $\sigma_3 > 0$ but small, the inverse is formally possible but practically fragile: small frequency errors may be amplified into large parameter errors. Thus the smallest singular value identifies the least observable parameter combination, while κ measures how strongly observational errors may be amplified in the inferred parameters. In practice, $\widetilde{J}$ is evaluated by central finite differences with relative step $\delta = 10^{-6}$.

3.2 Excitation filtering as a geometric statement

The inverse results concern what can be inferred once modal frequencies have been measured. Geometry does another job earlier in the chain: it limits which modes can be excited efficiently in the first place. For long-wavelength forcing this limitation is captured by the resonant absorption cross-section.

Proposition 1 (Breit–Wigner excitation filter) *Consider flexural mode n of a fluid-filled shell driven at resonance by an incident acoustic field of wavelength λ. Let ζ_{rad} and ζ_{str} denote the radiation and structural damping ratios of that mode. Then the resonant absorption cross-section is*

$$\sigma_{\mathrm{abs},n} = \frac{(2n+1)\lambda^2}{4\pi} \frac{4\zeta_{\mathrm{rad}}\zeta_{\mathrm{str}}}{(\zeta_{\mathrm{rad}} + \zeta_{\mathrm{str}})^2}. \tag{10}$$

In particular,

$$0 \le \sigma_{\mathrm{abs},n} \le \sigma_{\mathrm{max},n} := \frac{(2n+1)\lambda^2}{4\pi}, \tag{11}$$

with equality if and only if $\zeta_{\mathrm{rad}} = \zeta_{\mathrm{str}}$. *In the Rayleigh limit for flexural radiation,* $\zeta_{\mathrm{rad}} \propto (ka)^{2n+2}$; *hence if* $ka \ll 1$ *and* $\zeta_{\mathrm{rad}} \ll \zeta_{\mathrm{str}}$ *then*

$$\sigma_{\mathrm{abs},n} = \sigma_{\mathrm{max},n}\left(\frac{4\zeta_{\mathrm{rad}}}{\zeta_{\mathrm{str}}} + O\left(\frac{\zeta_{\mathrm{rad}}^2}{\zeta_{\mathrm{str}}^2}\right)\right), \tag{12}$$

so long-wavelength airborne forcing is strongly filtered even when the mode itself exists.

Proof. For a single resonant partial wave, reciprocity gives the standard Breit–Wigner absorption law [22],

$$\sigma_{\mathrm{abs},n} = \sigma_{\mathrm{max},n}\frac{4\Gamma_{\mathrm{rad}}\Gamma_{\mathrm{str}}}{(\Gamma_{\mathrm{rad}} + \Gamma_{\mathrm{str}})^2}, \tag{13}$$

where Γ_{rad} and Γ_{str} are the radiative and structural linewidths. Dividing by the modal natural frequency shows that the same expression holds in terms of damping ratios $\zeta_{\mathrm{rad}} = \Gamma_{\mathrm{rad}}/(2\omega_n)$ and $\zeta_{\mathrm{str}} = \Gamma_{\mathrm{str}}/(2\omega_n)$, which gives (10). Writing $x = \zeta_{\mathrm{rad}}/\zeta_{\mathrm{str}} \ge 0$ yields the efficiency factor $4x/(1+x)^2 \le 1$, with equality only at $x = 1$, proving (11). Finally, the Rayleigh-limit radiation resistance of the nth flexural mode scales as $(ka)^{2n+2}$, so $\zeta_{\mathrm{rad}} \propto (ka)^{2n+2}$. Expanding the efficiency factor for $x \ll 1$ gives $4x + O(x^2)$ and therefore (12). The mode may be present in the spectrum, but geometry has already reduced how much incident power reaches it. □

Scope of Proposition 1. Proposition 1 is stated for the spherical partial-wave expansion and is exact only in the spherical limit. For the oblate geometries considered here ($\varepsilon \approx 0.745$), the partial-wave cross-sections acquire correction terms involving oblate spheroidal harmonics. However, the qualitative conclusion — that low-order flexural modes couple inefficiently to airborne sound in the Rayleigh regime — is robust: the $(ka)^{2n}$ suppression reflects the general multipole scaling and is not an artefact of spherical symmetry. A quantitative treatment for oblate scattering would require the T-matrix formalism [22] and is beyond the present scope.

3.3 Equivalent-radius reductions and lost geometric dimension

Equivalent-sphere models replace the two-parameter geometry (a, c) by a single scalar descriptor, typically the equivalent radius

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{14}$$

or, more generally, some smooth scalar reduction $g(a, c)$. Such reductions can be entirely sensible for forward estimates of resonance magnitude or scaling, and they are used in [14, 15] for precisely that purpose. Their inverse implication is considerably less favourable.

Theorem 2 (Rank collapse under scalar geometric reduction) *Let a reduced forward model satisfy*

$$f_n(a, c, E; \mathbf{p}) = \phi_n\big(g(a, c), E; \mathbf{p}\big) \qquad \text{for all } n \in \mathcal{M}, \tag{15}$$

with $g : \mathbb{R}_+^2 \to \mathbb{R}_+$ *a smooth scalar function. Then the scaled Jacobian with respect to* $\boldsymbol{\theta} = (a, c, E)$ *satisfies*

$$\operatorname{rank}(\widetilde{J}) \le 2. \tag{16}$$

Equivalently, one independent geometric direction is unobservable from modal frequencies alone.

Proof. By the chain rule,

$$\frac{\partial f_n}{\partial a} = \frac{\partial \phi_n}{\partial g}\frac{\partial g}{\partial a}, \qquad \frac{\partial f_n}{\partial c} = \frac{\partial \phi_n}{\partial g}\frac{\partial g}{\partial c}. \tag{17}$$

Hence the a- and c-columns of the raw Jacobian are both scalar multiples of the same vector $\left(\partial\phi_{n_i}/\partial g\right)_{i=1}^{m}$, so the geometric block of J has rank at most 1. Appending the E-column can increase the rank by at most one, which gives $\operatorname{rank}(J) \le 2$.

Now write the scaled Jacobian as

$$\widetilde{J} = D_f^{-1} J D_\theta, \tag{18}$$

where $D_f = \operatorname{diag}(f_{n_1}, \dots, f_{n_m})$ and $D_\theta = \operatorname{diag}(a, c, E)$. Both diagonal matrices are nonsingular for physically admissible parameters, so left- and right-multiplication do not change rank. Therefore $\operatorname{rank}(\widetilde{J}) = \operatorname{rank}(J) \le 2$, as claimed. For the specific reduction $g = R_{\mathrm{eq}} = (a^2c)^{1/3}$ one obtains $\partial R_{\mathrm{eq}}/\partial a = (2/3)R_{\mathrm{eq}}/a$ and $\partial R_{\mathrm{eq}}/\partial c = (1/3)R_{\mathrm{eq}}/c$, so the scaled columns satisfy $\widetilde{J}[:, a] = 2\,\widetilde{J}[:, c]$ exactly. □

Theorem 2 is stronger than a complaint about poor conditioning. It states that the missing information is *structural*: no amount of experimental precision can recover a geometric direction that the model has collapsed into a single scalar. One may still estimate $g(a, c)$ and E, but not a and c separately.

3.4 Asphericity as an identifiability-lifting perturbation

To escape Theorem 2, the model must retain at least two genuinely independent geometric channels. For an oblate shell this occurs through curvature anisotropy. Writing the eccentricity as

$$\varepsilon = \sqrt{1 - c^2/a^2}, \tag{19}$$

the spherical limit is $\varepsilon = 0$, while $\varepsilon > 0$ introduces distinct meridional and circumferential curvature distributions. In the Ritz formulation detailed in Appendix A, mode n samples those curvature fields through mode-specific weighting functions. Schematically,

$$f_n(\varepsilon) = f_n^{(0)} + \alpha_n \varepsilon^2 + \beta_n \varepsilon^4 + \cdots, \tag{20}$$

where the coefficients $\alpha_n, \beta_n, \dots$ depend on the mode number because different mode shapes concentrate strain energy in different regions of the shell. Once those coefficients vary with n, the a- and c-columns of $\widetilde{J}$ need no longer remain proportional across modes.

Proposition 3 (Identifiability lifting by oblate asphericity) *Consider the oblate Ritz shell model with measured flexural modes $\mathcal{M} \supseteq \{2, 3, 4\}$ and inversion parameters $\boldsymbol{\theta} = (a, c, E)$. Then there exists a non-empty open set $U \subset \{\boldsymbol{\theta} \in \Theta : c < a\}$ on which*

$$\operatorname{rank}(\widetilde{J}(\boldsymbol{\theta})) = 3. \tag{21}$$

Thus oblate asphericity lifts the scalar-reduction rank collapse and restores local identifiability of (a, c, E).

Proof. The proof is by numerical construction at the canonical operating point, extended by continuity. For a fixed Ritz basis and a fixed measured-mode set, the frequencies $f_n(\boldsymbol{\theta})$ are smooth functions of (a, c, E) away from modal crossings, so the entries of $\widetilde{J}(\boldsymbol{\theta})$ are smooth as well. At the canonical oblate operating point

$$\boldsymbol{\theta}_* = \big(0.18\,\mathrm{m}, 0.12\,\mathrm{m}, 0.1\,\mathrm{MPa}\big), \tag{22}$$

the numerical study in [16] evaluates the scaled Jacobian by central finite differences and reports $\kappa(\widetilde{J}(\boldsymbol{\theta}_*)) = 69.4$ for the five-mode inverse using modes $n = 2, \ldots, 6$. A finite condition number requires all three singular values to be strictly positive; therefore $\sigma_3(\widetilde{J}(\boldsymbol{\theta}_*)) > 0$ and $\mathrm{rank}(\widetilde{J}(\boldsymbol{\theta}_*)) = 3$.

Singular values depend continuously on the matrix entries, hence continuously on $\boldsymbol{\theta}$. Since $\sigma_3(\widetilde{J}(\boldsymbol{\theta}_*))$ is strictly positive, there exists $\delta > 0$ such that $\sigma_3(\widetilde{J}(\boldsymbol{\theta})) > \sigma_3(\widetilde{J}(\boldsymbol{\theta}_*))/2 > 0$ whenever $\|\boldsymbol{\theta} - \boldsymbol{\theta}_*\| < \delta$. Intersecting that ball with the oblate set $\{c < a\}$ yields a non-empty open neighbourhood U on which $\mathrm{rank}(\widetilde{J}) = 3$. □

The result is existential rather than global. It does not say that every oblate shell is well conditioned. It says that oblate asphericity restores the *possibility* of independent recovery by breaking the exact proportionality imposed by scalar geometric reduction. Conditioning then becomes a quantitative matter rather than an algebraic impossibility.

3.5 Near-spherical asymptotics and the corrected spherical limit

The next question is how the restored identifiability behaves as the shell approaches spherical symmetry. The correct statement is subtler than the first draft of the capstone suggested. At the exact sphere, symmetry forces rank collapse and therefore $\sigma_3(0) = 0$. The previously reported non-zero intercept $\sigma_0 > 0$ was a low-order Ritz discretisation artefact, not a property of the continuum inverse problem. What survives is a regular even expansion in the curvature parameter.

Proposition 4 (Regular near-spherical asymptotics) *Consider a smooth one-parameter oblate family of Ritz shell models parameterised by the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$, with $\varepsilon = 0$ at the sphere. Then the smallest singular value of the scaled Jacobian satisfies*

$$\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4), \qquad \varepsilon \to 0, \tag{23}$$

with $\lambda_1 > 0$ for the canonical oblate family considered in [16, 19]. The analytic regularity of the eigenvalue perturbation is guaranteed by Kato's perturbation theory for linear operators [9]. In particular,

$$\sigma_3(0) = 0. \tag{24}$$

Proof. Reflection symmetry implies that replacing ε by $-\varepsilon$ describes the same oblate geometry. The stiffness and mass operators of the Ritz model are therefore even analytic functions of ε, so the frequency map and the scaled Jacobian admit expansions of the form

$$\widetilde{J}(\varepsilon) = \widetilde{J}_0 + \varepsilon^2 \widetilde{J}_2 + O(\varepsilon^4). \tag{25}$$

At $\varepsilon = 0$ the shell is spherical, so geometry enters only through a single radius and Theorem 2 applies. Hence $\mathrm{rank}(\widetilde{J}_0) \le 2$ and therefore $\sigma_3(0) = 0$.

To determine the leading power, consider the Gram matrix $A(\varepsilon) = \widetilde{J}(\varepsilon)^\top \widetilde{J}(\varepsilon)$, whose eigenvalues are $\mu_k = \sigma_k^2$. The even expansion of $\widetilde{J}$ gives

$$A(\varepsilon) = A_0 + \varepsilon^2 A_2 + O(\varepsilon^4), \tag{26}$$

where $A_0 = \widetilde{J}_0^\top \widetilde{J}_0$ and $A_2 = \widetilde{J}_0^\top \widetilde{J}_2 + \widetilde{J}_2^\top \widetilde{J}_0$. Let $\mathbf{v}_0 \in \ker(\widetilde{J}_0)$ be the unit null vector (guaranteed to exist since $\sigma_3(0) = 0$). The first-order eigenvalue perturbation of μ_3 vanishes:

$$\mathbf{v}_0^\top A_2\, \mathbf{v}_0 = \mathbf{v}_0^\top \big(\widetilde{J}_0^\top \widetilde{J}_2 + \widetilde{J}_2^\top \widetilde{J}_0\big)\mathbf{v}_0 = 0, \tag{27}$$

because $\widetilde{J}_0\, \mathbf{v}_0 = \mathbf{0}$. The eigenvalue $\mu_3(\varepsilon)$ therefore enters at second order in ε^2, i.e. at order ε^4. By the analytic perturbation theory of Kato [9], the eigenvalue branches of $A(\varepsilon)$ are themselves analytic in the perturbation parameter where they remain simple, so $\mu_3(\varepsilon) = \lambda_1^2 \varepsilon^4 + O(\varepsilon^6)$ and hence $\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4)$.

For the canonical oblate family, Proposition 3 gives $\sigma_3(\varepsilon) > 0$ on an open set of $\varepsilon > 0$, so the leading coefficient satisfies $\lambda_1 > 0$, yielding (23). The corrected near-spherical analysis therefore removes the earlier fictitious intercept $\sigma_0 > 0$: the apparent finite floor was a discretisation artefact of the low-order spherical limit, not a true continuum property. □

Corollary 5 *If $\sigma_1(\varepsilon) = \sigma_1(0) + O(\varepsilon^2)$ with $\sigma_1(0) > 0$, then*

$$\kappa(\widetilde{J}(\varepsilon)) = \frac{\sigma_1(0)}{\lambda_1}\, \varepsilon^{-2} + O(1) \qquad \text{as } \varepsilon \to 0. \tag{28}$$

The corollary has a simple interpretation. The exact sphere is singular, but the loss of identifiability is regular rather than exotic: the condition number grows quadratically as spherical symmetry is restored.

3.6 Forward adequacy and inverse adequacy are different questions

Reduced models are often judged by forward error alone. Let $\mathbf{f}_{\text{red}}$ denote a reduced model and $\mathbf{f}_{\text{full}}$ a geometry-resolving model. A natural forward metric is the relative frequency misfit

$$\mathcal{E}_{\text{fwd}} = \frac{\|\mathbf{f}_{\text{red}}(\boldsymbol{\theta}) - \mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}{\|\mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}. \tag{29}$$

An inverse metric, by contrast, must assess the conditioning of the parameter recovery map; locally, $\kappa(\widetilde{J})$ is the appropriate quantity. There is no theorem forcing small $\mathcal{E}_{\text{fwd}}$ to imply moderate κ. The Browntone sequence instead supplies explicit counterexamples.

Proposition 6 (Forward adequacy does not imply inverse adequacy) *There exist parameter domains on which a reduced equivalent-sphere model has small forward error while remaining unusable for joint recovery of (a, c, E). In particular, at the canonical abdominal operating point and using modes $n = 2, \ldots, 6$,*

$$\mathcal{E}_{\text{fwd}} < 0.1, \tag{30}$$

yet the equivalent-sphere model satisfies

$$\operatorname{rank}(\widetilde{J}_{\text{sphere}}) \le 2 \qquad \text{and} \qquad \kappa(\widetilde{J}_{\text{sphere}}) \approx 1.37 \times 10^{10}, \tag{31}$$

whereas the oblate Ritz model on the same modal set has

$$\operatorname{rank}(\widetilde{J}_{\text{oblate}}) = 3 \qquad \textit{and} \qquad \kappa(\widetilde{J}_{\text{oblate}}) = 69.4. \tag{32}$$

The mechanism is a structural sensitivity gap *along the kernel of the geometric reduction.*

Proof. At the canonical abdominal point, the equivalent-sphere and oblate Ritz models produce closely matched low-order spectra: the relative forward misfit for modes $n = 2, \ldots, 6$ is $\mathcal{E}_{\text{fwd}} = 0.0936 < 0.1$. Thus the reduced model is forward-adequate in the ordinary engineering sense of reproducing the dominant frequencies to within 10 %. By Theorem 2, however, the equivalent-sphere inverse map is structurally rank deficient, so $\operatorname{rank}(\widetilde{J}_{\text{sphere}}) \leq 2$. The finite-difference evaluation reported in [16] gives the associated numerical condition number $\kappa(\widetilde{J}_{\text{sphere}}) \approx 1.37\times10^{10}$.

On the same parameter domain, Proposition 3 shows that the oblate Ritz model has full rank on an open neighbourhood of the canonical point, and [16] gives the concrete value $\kappa(\widetilde{J}_{\text{oblate}}) = 69.4$ for the five-mode inverse [16].

The structural reason why forward and inverse adequacy decouple is made transparent by the kernel of the projection $\pi : (a, c, E) \mapsto (R_{\text{eq}}, E)$. Let $\mathbf{v} \in \ker D\pi(\boldsymbol{\theta}_0)$ with $\|\mathbf{v}\| = 1$; this is the direction that trades a against c while preserving $R_{\text{eq}} = (a^2c)^{1/3}$. Then

$$\widetilde{J}_{\text{sphere}}\,\mathbf{v} = DG(\pi(\boldsymbol{\theta}_0))\ \underbrace{D\pi(\boldsymbol{\theta}_0)\,\mathbf{v}}_{=\mathbf{0}} = \mathbf{0}, \tag{33}$$

so the reduced model is *exactly insensitive* to perturbations along $\mathbf{v}$. By contrast, $\operatorname{rank}(\widetilde{J}_{\text{oblate}}) = 3$ guarantees $\widetilde{J}_{\text{oblate}}\,\mathbf{v} \neq \mathbf{0}$: the full model detects parameter changes to which the reduced model is structurally blind.

Forward adequacy (29) constrains the *values* of the two frequency vectors to be close; it places no constraint on how their *derivatives* compare along the lost direction, because the forward error lives in the image space $\mathbb{R}^m$ whereas the inverse problem concerns the preimage structure in Θ. A model can therefore reproduce measured frequencies well yet be wholly unable to distinguish the parameter combinations that produced them. □

Remark 7 (Watermelon model as supporting example) *An instructive companion example is provided by the watermelon model in [15]: there the equivalent-sphere reduction succeeds precisely because geometry is fixed independently and the inverse target is effectively one-dimensional. This separation parallels the model-updating literature's long-standing distinction between residual minimisation and parameter observability [23, 3]. Forward adequacy is not false; it is simply insufficient for multivariate spectral inversion.*

Remark 8 (Near-spherical divergence of the adequacy gap) *In the near-spherical regime, the coexistence of forward adequacy and inverse inadequacy becomes increasingly extreme. The forward error shrinks as the shell approaches spherical symmetry because the equivalent-sphere approximation becomes exact at $\varepsilon = 0$. But the condition number diverges by Corollary 5: $\kappa(\widetilde{J}) \sim (\sigma_1(0)/\lambda_1)\,\varepsilon^{-2}$ as $\varepsilon \longrightarrow 0$. The ratio of inverse difficulty to forward discrepancy therefore grows without bound: the sphere limit is simultaneously the most forward-adequate and the most inverse-inadequate configuration. Figure 2 illustrates this divergence numerically.*

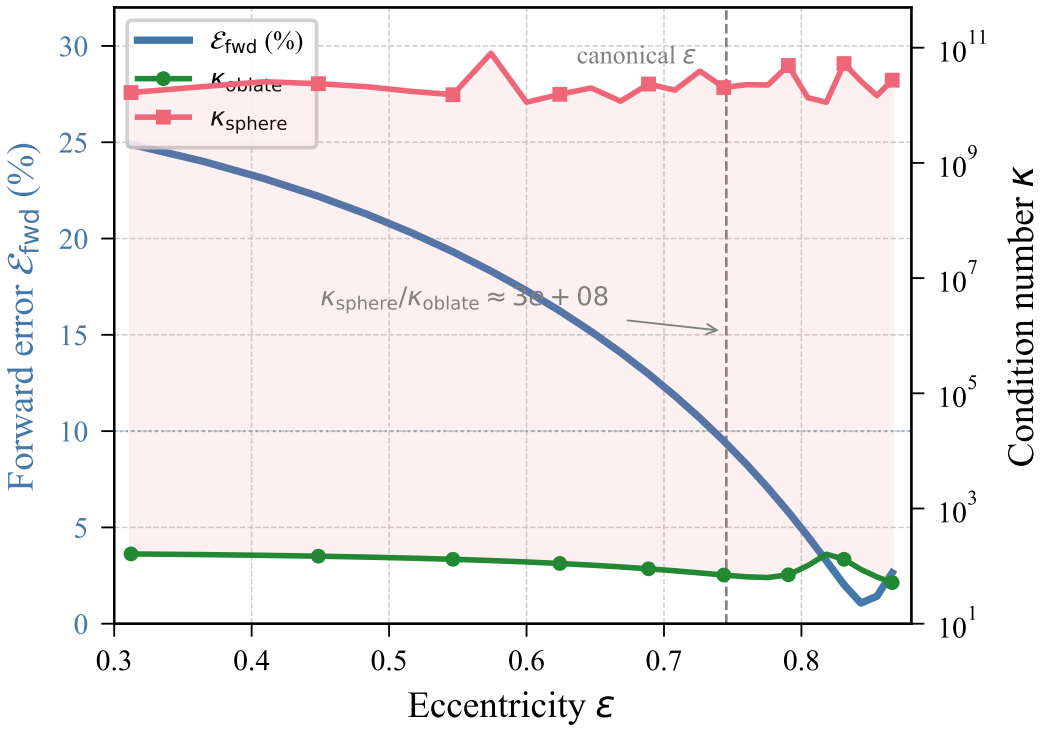

Figure 2: The forward-inverse adequacy gap (Proposition 6). The forward error $\mathcal{E}_{\text{fwd}}$ (blue, left axis) quantifies how well the equivalent-sphere model reproduces oblate Ritz frequencies; the condition number κ (right axis) quantifies how usable each model is for parameter recovery. The oblate Ritz model (green) maintains moderate κ across all eccentricities, while the equivalent-sphere model (red) is structurally rank deficient everywhere. Both models share the same forward-error curve, yet their inverse conditioning differs by eight orders of magnitude. The canonical operating point ($\varepsilon = 0.745$, vertical dashed line) sits in a region where $\mathcal{E}_{\text{fwd}} < 10\%$ and $\kappa_{\text{sphere}}/\kappa_{\text{oblate}} \approx 2 \times 10^8$.

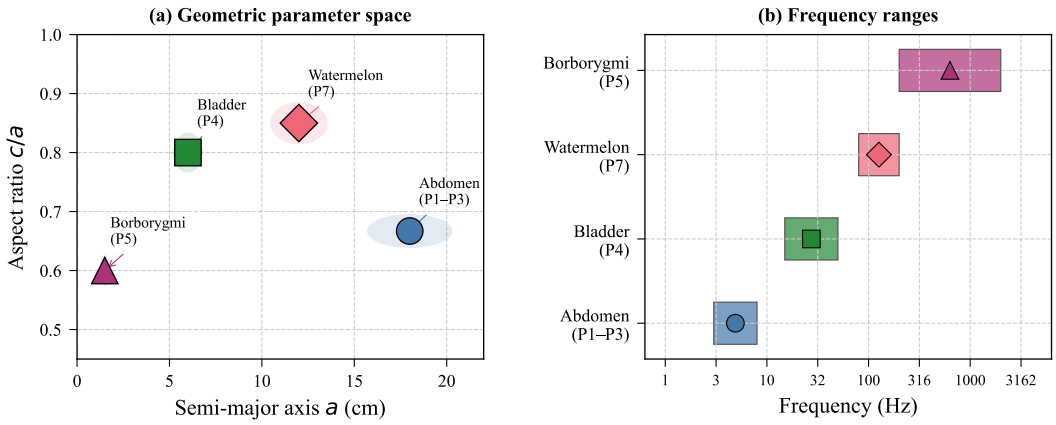

Figure 3: Four application domains in geometric parameter space, showing the range of shell geometries and characteristic frequencies across the programme.

Taken together, Theorem 2, Proposition 3, Proposition 4, and Proposition 6 define the capstone thesis in mathematical form. Scalar geometric reductions collapse one inverse direction; oblate curvature can restore it; the restoration vanishes quadratically at the exact sphere; and forward success does not justify inverse use. The results section tests these formal statements against the cross-paper evidence.

4 Results

We now test the four formal results against the cross-application record assembled in [14, 17, 13, 15, 16]. The point of the comparison is not that every application solves the same inverse problem. Instead, some cases are dominated by excitation filtering, some by spectral organisation, and some by parameter recovery. What persists across them is that geometry decides which of those questions can be asked sensibly in the first place. Table 1 collects the principal application-level evidence used below.

Table 1 already hints at the main result. The abdominal, bladder, watermelon, and borborygmi studies occupy different frequency bands and involve different observables, yet they all distinguish between a forward spectral pattern and the stronger claim that parameters can be recovered independently from that pattern. That distinction becomes explicit once the four formal results are read as a sequence rather than as isolated claims.

4.1 Rank collapse under scalar geometric reduction

The theorem is the most stringent mathematically. If the flexural spectrum depends on the geometry only through the scalar reduction

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{34}$$

then the two geometric sensitivity directions cannot remain independent. By the chain rule, every modal derivative with respect to the equatorial semi-axis a is proportional to the corresponding derivative with respect to the polar semi-axis c. In the scaled Jacobian $\widetilde{J}$, the two geometry columns

are therefore collinear. At the sphere this becomes physically transparent: a small perturbation in a and a small perturbation in c both act only by changing the same radius, so the geometry columns become proportional and one geometric direction disappears from the inverse map.

The identifiability analysis in [16] made that loss concrete for the canonical abdominal parameter set. The reported condition number for the equivalent-sphere inverse problem is

$$\kappa_{\text{sphere}} \approx 1.37 \times 10^{10}, \tag{35}$$

numerically finite only because the derivatives were estimated with a finite difference rather than evaluated symbolically [16]. The true statement is stronger than "poorly conditioned": the model architecture has already removed one independent geometric degree of freedom before any data are fitted.

The cross-application evidence supports this interpretation rather than weakens it. In the watermelon study of [15], the equivalent-sphere model works because the inverse task is deliberately narrow: the fruit geometry is measured separately, and the frequency is used to infer one effective stiffness scale E_{rind} [15]. In the constrained-bubble analysis of [13], the borborygmi model is similarly successful only because the dominant unknown is effectively the gas-pocket size or mode family, not the simultaneous recovery of two shape axes and a wall modulus [13]. These are not counterexamples to rank deficiency. They are examples of problems that avoid it by reducing the inverse target. Once one asks for joint recovery of (a, c, E) from spectra alone, the equivalent-radius reduction has already discarded too much of the available information.

4.2 Identifiability lifting by oblate asphericity

The next result asks whether this degeneracy is merely diagnosed or actually repairable. The evidence says that it is repairable, but only when the model retains genuine curvature information. In the oblate Ritz family analysed in [16], we may sweep continuously from the spherical limit toward the canonical abdominal geometry by decreasing the aspect ratio $\zeta = c/a$ from near unity to $\zeta = 0.667$, equivalently increasing the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$ from almost zero to the canonical oblate value $\varepsilon = 0.745$ [16].

Two comparisons matter. The first is the broad conceptual one: the condition number drops from the 1×10^{10}-scale equivalent-sphere pathology to

$$\kappa_{\text{oblate}} = 69.4 \tag{36}$$

once the geometry is treated as oblate rather than spherical. The second is the stricter like-for-like comparison within the same oblate Ritz model: the conditioning improves from the near-spherical floor to the canonical oblate point without changing the inverse variables or the measured mode set. In both comparisons the message is the same. Asphericity is not merely a geometric detail; it reintroduces a second geometric sensitivity channel.

The physical reason is mode-dependent curvature sampling. The lowest abdominal flexural mode already sits at

$$f_2 = 3.95\,\text{Hz}, \tag{37}$$

so the improvement in identifiability does not arise because the resonance is dragged into a more convenient frequency band. It arises because the higher flexural modes do *not* sample the oblate

shell in the same way. Meridional and circumferential curvature changes are weighted differently by different mode shapes, so the a- and c-columns of $\widetilde{J}$ cease to be parallel. The bladder study provides a useful supporting contrast here. Its fill-dependent U-shaped curve shows that geometry and stiffness can compete strongly in the forward spectrum [17]; the identifiability analysis in [16] shows what is additionally needed for inversion, namely that different modes must separate those effects rather than merely combine them.

This is also where the abdominal coupling result matters conceptually. In the abdominal model of [14], the same oblate geometry that yields the canonical $f_2 = 3.95\,\mathrm{Hz}$ also produces a severe excitation asymmetry, with a reported mechanical-to-airborne displacement ratio $R \approx 3.3 \times 10^4$ in the canonical comparison [14]. The same geometry therefore controls both whether a mode is efficiently excited and whether it is diagnostically informative. This encapsulates the capstone thesis: geometry controls both the efficiency of excitation and the information content of the measured response.

4.3 Regular near-spherical asymptotics

The third formal result concerns the approach to the sphere. A tempting but incorrect intuition is that the near-spherical oblate inverse problem should retain a finite condition-number floor because early low-order Ritz fits appeared to give a positive intercept σ_0. The corrected analysis is stricter. At the exact sphere, restored symmetry forces rank collapse, so $\sigma_3(0) = 0$. What remains is the regular even expansion

$$\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4), \tag{38}$$

with $\lambda_1 > 0$ for the canonical oblate family. The previously reported finite floor was therefore a discretisation artefact of the low-order Ritz spherical limit, not a property of the continuum inverse problem.

The resulting classification has three numerically distinct classes. The first class is the *singular sphere*: the equivalent-sphere reduction has $\kappa_{\mathrm{sphere}} \approx 1.37 \times 10^{10}$ because the inverse map is rank deficient. The second is the *regular oblate* class: the curvature-resolving oblate family is full rank on the tested oblate branch and, in particular, on a non-empty open set around the canonical oblate operating point, with conditioning that deteriorates like ε^{-2} as the sphere is approached and improves to 69.4 at the canonical oblate point. The third is the *non-lifting prolate* class: breaking spherical symmetry in the wrong geometric direction does not help much, and the condition number stays in the range

$$\kappa_{\mathrm{prolate}} \approx 561\text{–}729 \tag{39}$$

across the tested prolate branch [16, 19].

This three-class picture matters because it replaces a vague slogan — "non-spherical is better" — with a more precise statement. Some departures from the sphere restore useful inverse information; some do not; and some model reductions remain singular regardless of numerical precision. The borborygmi and bladder papers again provide supporting context. Both show that spectra in soft, fluid-loaded biological systems can change strongly with geometry, fill state, or confinement [17, 13]. The near-spherical analysis tells us when such spectral variation should also be expected to carry independent inverse information. Those are related questions, but they are not the same question.

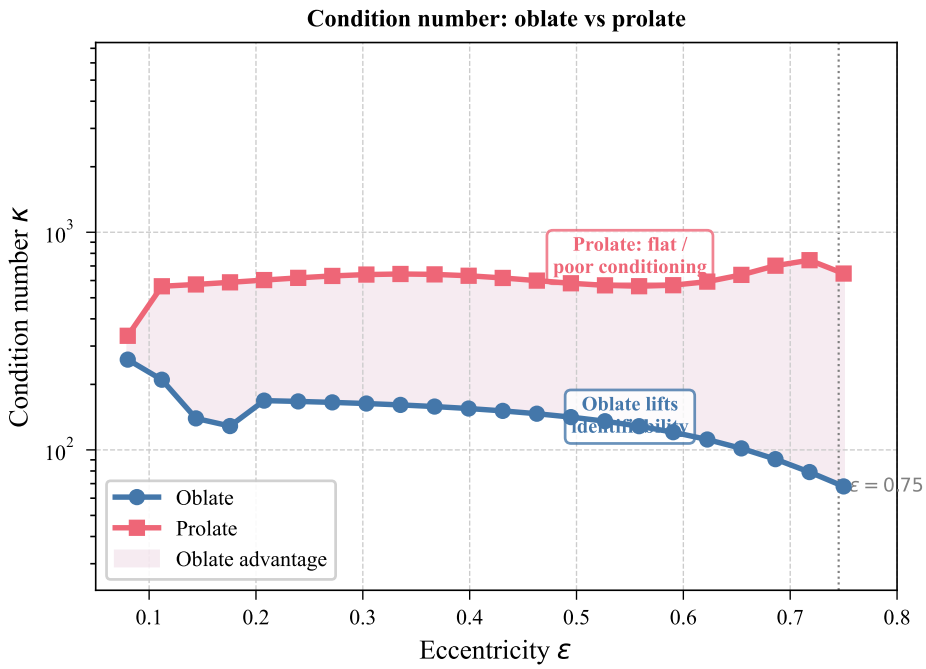

Figure 4: Condition number κ as a function of asphericity ε for oblate and prolate families. The oblate branch shows systematic improvement; the prolate branch remains flat.

4.4 Forward adequacy does not imply inverse adequacy

The fourth formal result is the most practical, and the watermelon analysis in [15] supplies the most useful cautionary example. The equivalent-sphere watermelon model predicts tap-tone frequencies with useful accuracy over the relevant stiffness range: f_2 spans 46.4 Hz to 203.2 Hz, and at the canonical reference point the model gives $f_2 = 89.9$ Hz with a monotonic relation between f_2^2 and the effective rind modulus E_{rind} [15]. For that task the reduced model is appropriate. Geometry is measured independently; the inverse target is essentially one-dimensional; and the model therefore serves as a practical forward predictor and calibration law.

The difficulty arises when that forward success is promoted into a general inversion claim. If the same equivalent-sphere architecture is asked to recover (a, c, E) jointly from modal data, the condition number returns to the 1×10^{10} scale, because the geometry has again been collapsed into a single radius. The curvature-resolving oblate model, by contrast, yields broadly similar low-order frequencies while restoring a usable inverse map with

$$\kappa = 69.4. \tag{40}$$

Small forward spectral error and good inverse conditioning are related only loosely, and sometimes not at all.

The other applications reinforce the same lesson from different angles. The bladder model predicts a clear forward phenomenon — a minimum $f_2 = 13.5$ Hz at 222 mL — and a clear coupling phenomenon — $\mathcal{R}_{\text{bladder}} \approx 6{,}400$ in favour of mechanical excitation [17]. Neither result, by itself, certifies that a joint inverse problem for geometry and material parameters would be well conditioned. The borborygmi model similarly supports narrow inferences, such as mapping a peak

near 340 Hz to a pocket volume of order 3 mL, without claiming that bowel geometry and wall elasticity can be reconstructed uniquely from the sound alone [13]. Forward adequacy is often enough for diagnosis, classification, or design. It does not obviate the need to assess identifiability.

Taken together, the four formal results survive the cross-application synthesis. Equivalent-radius reductions are structurally rank deficient; oblate curvature can lift that defect; the lifting vanishes regularly at the exact sphere; and forward predictive success does not guarantee inverse usefulness. That is why the same research programme can contain, without contradiction, a successful equivalent-sphere watermelon model and a strong warning against equivalent-sphere inversion. The models answer different questions. Geometry determines which inferences are defensible.

5 Discussion

The capstone claim of this paper is not simply that geometry matters. That would be a qualitative observation rather than a mechanistic one. The stronger conclusion is that geometry enters the problem of fluid-filled viscoelastic shells in three logically distinct ways, and that confusing those roles obscures both the forward physics and the inverse problem. Geometry filters how forcing is projected onto admissible deformation fields; geometry organises the modal spectrum through curvature, symmetry, and fluid loading; and geometry determines whether that spectrum contains enough independent information to recover the parameters that produced it. The four formal results of Section 3 are best read as consequences of this triple role, rather than as disconnected observations assembled from a sequence of applications.

5.1 Forward adequacy is not inverse adequacy

Of the four formal results, the deepest is the fourth: forward adequacy does not imply inverse adequacy. A reduced model may reproduce the dominant resonance frequencies of a shell to within a tolerance that is entirely acceptable for classification, screening, or preliminary design, yet remain unusable for parameter recovery because the inverse map is ill-conditioned or effectively singular. The present programme suggests that this decoupling is not an incidental numerical inconvenience. It is geometry-mediated.

Equivalent-radius reductions explain the practical appeal. If the observable of interest is a small set of low-order frequencies, a scalar geometric surrogate can often preserve the leading scale of the answer. Such models compress the problem without obviously destroying the forward spectrum, and that economy is precisely why they remain attractive in practice. What they do not preserve is *separability*. Once independent curvature information has been collapsed into a single scalar descriptor, distinct combinations of geometry and material stiffness can generate nearly indistinguishable spectra. The forward error may remain small while the inverse uncertainty becomes enormous. A model may therefore look persuasive in prediction and still be defective in inference.

This distinction is the most important conceptual result to emerge from the Browntone sequence. The existence of a resonance is not enough to establish an effective excitation pathway [14, 21]. Nor is the existence of a good spectral fit enough to establish a recoverable parameter map [15, 16]. These are parallel lessons. In both cases, geometry selects what can be seen. On the

excitation side, it selects which forcing fields project efficiently onto the mode family of interest. On the inverse side, it selects which parameter directions remain distinguishable once those modes are measured. A reduced model that is useful in forward prediction may therefore be unstable for inversion for reasons that are not accidental but structural.

5.2 Why geometry does three jobs

The first job of geometry is filtering. The original brown-note problem made this clear. A fluid-filled oblate shell can support a genuine low-frequency flexural mode near 4 Hz, yet airborne infrasound couples to that mode vanishingly weakly because the long-wavelength forcing field is itself filtered by the geometry of the enclosure [14]. The same mode is far more accessible to mechanical excitation because whole-body vibration is not subject to the same spatial-harmonic mismatch [21]. Geometry therefore acts as an impedance and projection filter between source and shell response. A resonance may exist without constituting a practical pathway.

The second job is organisation. Fluid-filled shells do not present arbitrary collections of frequencies. Curvature, aspect ratio, internal loading, and wall compliance sort the admissible motions into structured spectral families. In the abdominal problem, this distinguishes the low-frequency flexural modes from the much higher breathing mode dominated by fluid compressibility [14]. In the bladder, changing fill state alters both size and effective stiffness, producing a resonance trajectory that is not captured by a single monotonic size law [17]. In the gas-pocket and borborygmi papers, confinement and cavity geometry organise a different but related band of observable oscillations [18, 13]. Across applications, geometry is the principal organiser of what the spectrum is allowed to look like before any inversion begins.

Of the three geometric roles, excitation filtering and inverse identifiability are formalised as propositions and theorems; spectral organisation is supported by cross-application synthesis rather than by a comparably general formal result.

The third job is inversion, and this is the capstone contribution. Geometry governs which parameter directions are distinguishable in spectral space. When symmetry is high, mode sensitivities tend to align and the scaled Jacobian loses rank or becomes badly conditioned. When symmetry is broken in a way that creates genuinely different curvature sampling across modes, the sensitivity columns separate and inversion becomes possible. The rank-deficiency theorem, the identifiability-lifting proposition, and the near-spherical asymptotic proposition are all different ways of stating that fact. Geometry is not merely an input to the forward problem. It is the source of the information one hopes to recover from the spectrum.

5.3 Oblate–prolate asymmetry is a selectivity result

It would be tempting to interpret the inverse results as evidence that any departure from spherical symmetry is beneficial. The present evidence does not support so universal a reading. The oblate–prolate comparison instead suggests a more selective conclusion: some symmetry breakings produce useful modal differentiation, whereas others do not. Oblate geometries appear especially effective because they induce mode-dependent curvature sampling that decorrelates the parameter sensitivities. Prolate geometries, at least in the regimes so far examined, need not do so.

This matters because it changes how reduced models should be judged. If the aim is only to capture the leading frequency scale, an equivalent sphere may remain an entirely sensible first

approximation, and oblate and prolate corrections may appear equally modest at the level of forward error. If the aim is parameter recovery, however, the choice of reduced geometry becomes part of the inverse design problem. A reduction that preserves mean size while erasing curvature contrast may destroy identifiability; a reduction that retains the wrong kind of asphericity may improve very little. The asymmetry is therefore a selectivity result, not a universality result. Not all symmetry breaking helps, and reduced models should be assessed by the sensitivity structure they preserve, not by forward residuals alone.

That conclusion has broader methodological force. In many practical inverse problems, model reduction is justified *a posteriori* by a small mismatch in the predicted observables. The present results suggest a stricter criterion. A reduction is acceptable only if it preserves the geometry-induced information structure required by the inference task. That is a harder requirement, but it is the more appropriate one.

5.4 What the cross-application evidence establishes

Taken individually, the Browntone studies addressed local questions. Could a human abdominal cavity possess a low-frequency flexural resonance? Could a tissue-constrained gas pocket radiate in the audible band? How does bladder fill state alter resonance? Can tap-tone data support inversion in a watermelon? Does asphericity lift inverse degeneracy? The capstone claim is that these are not merely isolated case studies. They instantiate a common mechanism in which geometry governs forcing pathways, spectral structure, and identifiability in tandem.

The evidence, however, is not uniform, and it should not be presented as though it were. The filtering evidence is clearest in the abdominal, gas-pocket, and sub-bass studies: excitation is geometry selective, and the same resonance can be practically invisible to one forcing pathway yet accessible to another [14, 18, 21]. The organisational evidence is clearest in the scaling, bladder, and borborygmi studies: geometric scaling, fill state, and confinement sort the spectrum into coherent families [20, 17, 13]. The inverse evidence is clearest in the watermelon and Kac-style studies, because they expose the tension between a satisfactory forward fit and a recoverable parameter map [15, 16]. What the programme establishes, then, is not merely a collection of isolated case studies but a coherent synthesis: apparently disparate vibroacoustic phenomena in fluid-filled shells can be read as manifestations of one geometry-mediated structure.

5.5 Limitations

Several limitations should be stated plainly. First, the present theory relies on a Ritz approximation. That is a reasonable and economical way to capture the low-order modal structure of smooth axisymmetric shells, but it remains an approximation. A richer basis, higher-order shell theory, or full finite-element verification would be desirable, especially in parameter regimes where modal veering or stronger fluid–structure interaction becomes important.

Secondly, the kinematics are intentionally reduced. The low-dimensional shell description isolates the dominant meridional and radial content required for the present formal results, but it cannot represent the full deformation space of a real viscoelastic enclosure. This simplification is analytically productive and predictively limiting in equal measure.

Thirdly, the capstone is theoretical rather than experimental. No dedicated phantom measurements or in vivo data have yet been used to test the inverse claims directly. The present

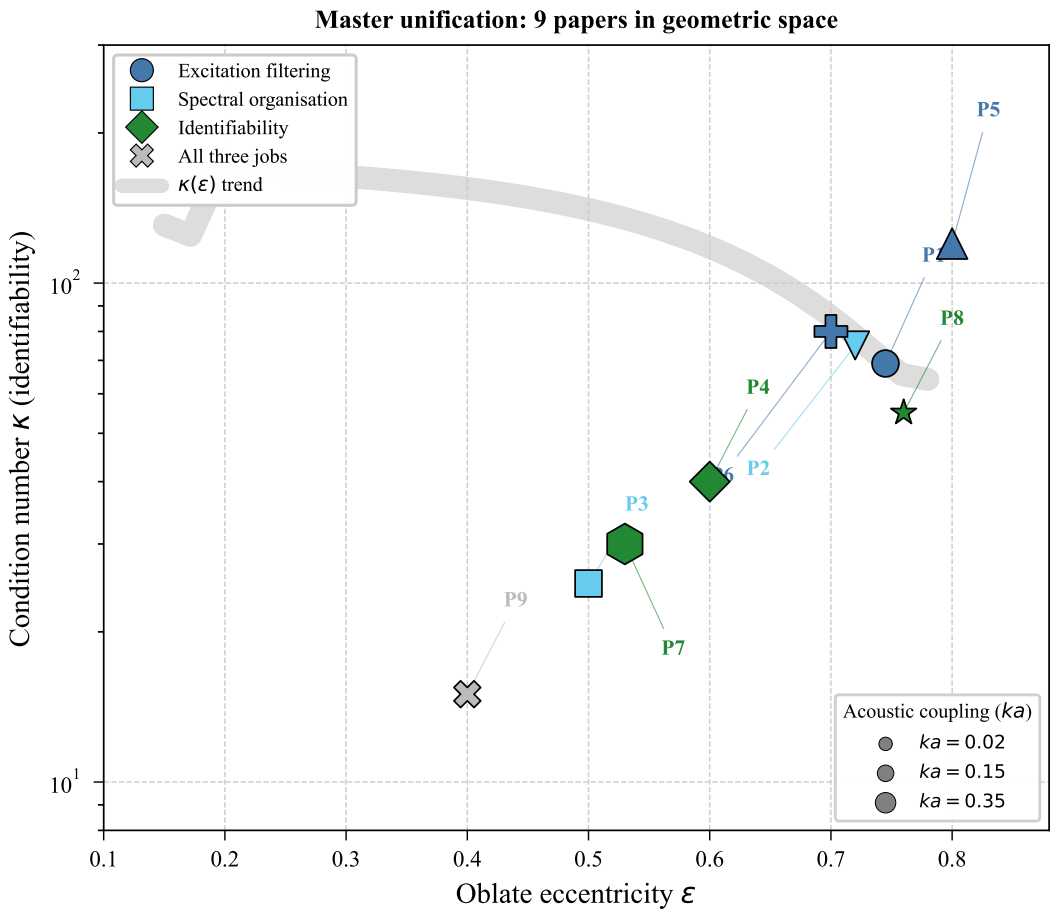

Figure 5: All nine companion papers mapped into a common eccentricity–conditioning space, illustrating the three geometric roles across applications.

conclusions should therefore be read as analytically supported and cross-application consistent, rather than experimentally closed.

Fourthly, the model class is intentionally narrow: thin elastic shells, single-fluid filling, small deformations, and linear response. Real biological and engineering enclosures may be layered, anisotropic, prestressed, spatially heterogeneous, or only approximately shell-like. Likewise, the surrounding supports and boundary conditions in vivo are more complicated than those represented here. Finally, the analysis remains confined to the linear regime and therefore excludes contact, finite strain, geometric stiffening, and other nonlinear effects that may become important in some applications.

These limitations do not negate the present argument, but they do mark its proper scope. The four formal results concern the geometric structure of spectral inference in a deliberately clean model class. Whether they survive, and in what modified form, under greater constitutive and geometric realism is precisely the next question.

5.6 Future directions

The most immediate next step is experimental validation. Phantom studies with controlled oblate and prolate shells would allow direct measurement of modal frequencies, damping, and inverse conditioning under known geometry and material properties. Such experiments matter because the capstone claims concern *parameter recovery*, not merely forward resonance. It is not enough to observe the correct frequency shift. One must test whether the recovered parameters are stable, repeatable, and geometry dependent in the predicted way. If that programme succeeds, in vivo measurements would provide the more difficult but more consequential test of whether the same structure persists in biological variability.

A second direction is theoretical extension beyond the linear thin-shell setting. Layered walls, anisotropy, prestress, nonlinear kinematics, and more realistic fluid–structure coupling may all alter the conditioning landscape. A particularly important question is which features of the present picture are robust consequences of symmetry and curvature, and which depend more delicately on the present reduced shell model.

A third direction is application. The same spectral-mechanics template may be useful for other compliant enclosures in biomechanics, soft robotics, membrane-lined vessels, industrial casings, and postharvest systems. In noise, vibration, and harshness, non-destructive testing, and biomedical sensing, one often seeks to infer internal state from sparse spectral observables. The present work suggests that success in such tasks may depend less on constitutive ornament and more on understanding which geometric features create distinguishable modal signatures.

Finally, the formal results deserve extension to richer model classes. The rank-deficiency and identifiability-lifting claims should be carried to higher order shell theories, the near-spherical asymptotics should be re-derived for layered and non-axisymmetric geometries, and the oblate–prolate selectivity should be characterised beyond the regimes so far examined. If the programme began with an unconventional motivating question, it now defines a tractable set of mathematical problems about symmetry, curvature, and information in shell spectra.

6 Conclusion

This paper has argued that the Browntone programme is best understood not as a collection of disparate applications, but as a coherent theory of geometry in fluid-filled compliant shells. The central thesis is that geometry does three jobs. It filters how external forcing couples into admissible shell motions; it organises the modal spectrum through symmetry, curvature, and fluid loading; and it determines whether the spectrum can be inverted to recover the parameters that generated it. That thesis emerged first from the abdominal and sub-bass problems, broadened through bladder, gas-pocket, scaling-law, and borborygmi studies, and reached its sharpest form in the inverse analyses of watermelon acoustics and organ-shape identifiability [14, 18, 20, 17, 13, 21, 15, 16].

Within that framework, four formal results have now been established. The first shows that equivalent-radius reductions of non-spherical fluid-filled shells are generically rank deficient, because they collapse independent geometric information into a single size parameter. The second shows that oblate asphericity can lift that deficiency by allowing different modes to sample curvature differently. The third shows that the transition to the spherical limit is a regular asymptotic problem with $\sigma_3(\varepsilon) = \lambda_1\varepsilon^2 + O(\varepsilon^4)$, so the exact sphere is singular and the previously

reported finite floor was a discretisation artefact. The fourth constructs a counterexample showing that forward adequacy does not imply inverse adequacy. A model may predict frequencies with apparently satisfactory accuracy while remaining unsuitable for parameter recovery.

The cross-application evidence suggests, moreover, that identifiability lifting is geometry selective rather than universal. Oblate perturbations appear able to generate the modal differentiation needed for stable inversion, whereas prolate perturbations need not do so. Reduced models for spectral inference should therefore be judged not only by forward accuracy, but by whether they preserve the geometry-induced sensitivity structure required by the inverse task.

We have been careful not to overstate what has been established. The capstone is analytical, not experimental. The shell models are linear, thin, and intentionally reduced. Even so, the present paper now places the central inverse claims on theorem and proposition level within that reduced class. A sufficiently consistent picture has therefore emerged to support a broader claim: spectral mechanics in fluid-filled compliant enclosures can be organised around geometry as an information-bearing structure, rather than treated as a collection of isolated frequency calculations.

That perspective offers a template for future work in biomechanics, biomedical sensing, non-destructive testing, and industrial noise, vibration, and harshness. Wherever one seeks to infer internal state, material condition, or geometry from sparse resonance data, the same question arises: which aspects of shape are merely descriptive, and which make the inverse problem possible? The present programme suggests that this is the right question to ask first.

In compressed form, the argument is this: geometry provides the organising principle of the programme. The next tasks are therefore clear. The formal results should be tested experimentally in phantoms and, where feasible, in vivo; the framework should be extended to nonlinear and more realistic shell models; and the present proofs should be generalised beyond the reduced axisymmetric class. Those steps would not change the direction of the present paper. They would substantially strengthen the analytical case advanced here.

Reduced models should be judged by the sensitivity structure they preserve, not by forward error alone.

A Oblate Spheroidal Shell Ritz Model

This appendix collects the Rayleigh–Ritz formulation used throughout the companion papers and the present capstone. The model follows the classical shell-acoustics tradition [10, 7] with a Love–Kirchhoff thin-shell assumption.

A.1 Coordinate system and trial functions

The undeformed shell midsurface is an oblate spheroid with semi-major axis a and semi-minor axis c ($c < a$). Define the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$ and the semifocal distance $d = a\varepsilon$. In oblate spheroidal coordinates (ξ, η, φ), the shell surface sits at $\xi_0 = c/d$. Here $\eta = \cos\theta$ ranges over $[-1, 1]$ with θ the polar angle, and $\varphi \in [0, 2\pi)$ is the azimuthal angle. The metric function

$$G(\eta) = a^2\eta^2 + c^2(1 - \eta^2) \tag{41}$$

determines the position-dependent curvature. The surface element is $\mathrm{d}S = 2\pi\, a\sqrt{G}\,\mathrm{d}\eta$.

For each circumferential mode number $n \geq 2$, the trial displacement field is a two-degree-of-freedom (2-DOF) Ritz ansatz:

$$w(\eta) = \beta\, P_n(\eta), \qquad u(\theta) = \alpha\, \sin\theta\, P_n'(\cos\theta), \tag{42}$$

where w is the normal (radial) displacement, u is the tangential (meridional) displacement, P_n denotes the Legendre polynomial of degree n, and β, α are the Ritz amplitudes forming the generalised coordinate vector $\mathbf{q} = (\beta, \alpha)^\top$. Legendre polynomials are smooth on $[-1, 1]$ and regular at both poles, so the trial functions are admissible for a free (unconstrained) shell without requiring any additional boundary conditions.

A.2 Energy functionals and eigenvalue problem

The stiffness matrix $\mathbf{K}$ receives three contributions, all integrated over the shell surface via Gauss–Legendre quadrature (200 nodes by default):

1. *Membrane stiffness.* With extensional rigidity $C_m = Eh/(1-\nu^2)$, the membrane strain energy arises from meridional and circumferential strains that depend on P_n, P_n', and the spatially varying metric (41).

2. *Bending stiffness.* With flexural rigidity $D_f = Eh^3/[12(1-\nu^2)]$, the curvature-change energy involves P_n'' and the derivatives of the metric function, weighted by $\sqrt{G}$.

3. *Geometric prestress.* The intra-abdominal pressure P_{iap} induces a membrane stress resultant $N_1 = P_{\text{iap}}\, R_2/2$, where $R_2(\eta) = a\sqrt{G}/c$ is the circumferential radius of curvature. This contributes a geometric stiffness term (tension stiffening) that acts on the normal DOF only.

The mass matrix $\mathbf{M}$ has two contributions:

1. *Shell inertia.* The structural mass per unit area is $\rho_w h$, integrated against P_n^2 for the normal DOF and $\sin^2\theta\,(P_n')^2$ for the tangential DOF.

2. *Fluid added mass.* The interior fluid loading is computed from the Laplace problem in oblate spheroidal coordinates using Legendre functions $P_n(i\xi)$ on the imaginary argument, following Junger and Feit [7]. This yields a scalar added-mass contribution to the normal DOF. Near the spherical limit ($c/a > 0.98$), the oblate harmonic solution is blended with the Lamb sphere formula via a C^2-continuous interpolant to avoid numerical singularity in the coordinate transformation.

Stationarity of the Rayleigh quotient yields the 2×2 generalised eigenvalue problem

$$\mathbf{K}\mathbf{q} = \omega^2 \mathbf{M}\mathbf{q}, \tag{43}$$

from which the natural frequency of mode n is extracted as the lowest positive eigenvalue. Solving (43) independently for each n produces the flexural spectrum $\{f_n\}_{n\geq 2}$ used throughout this paper.

A.3 Free boundary conditions and rigid-body modes

The shell is modelled as free (unconstrained). No displacement or rotation boundary conditions are imposed. For $n = 1$, the deformation is a rigid-body translation of the shell; the fluid adds mass but provides no restoring force, so $\omega_1 = 0$. The present paper considers only $n \geq 2$, for which the restoring force arises from shell elasticity and prestress.

A.4 Convergence of the Ritz basis

The 2-DOF trial (42) can be enriched by retaining additional Legendre terms of the same parity:

$$w(\eta) = \sum_{i=0}^{N-1} \beta_i \, P_{n+2i}(\eta), \qquad u(\theta) = \sum_{j=0}^{N-1} \alpha_j \, \sin\theta \, P'_{n+2j}(\cos\theta), \tag{44}$$

giving a $2N$-DOF problem. Note that in the enriched basis, the fluid added mass is computed per-degree using oblate spheroidal harmonics for each individual Legendre order $n + 2i$; off-diagonal fluid coupling between different Legendre orders is neglected.

For the canonical abdominal parameters (Table 1) and the $n = 2$ mode, the convergence study in [14] yields the results summarised in Table 2.

Three properties confirm adequacy of the 2-DOF basis. First, frequencies decrease monotonically with N, consistent with the Rayleigh–Ritz upper-bound property. Second, the 2-DOF model ($N = 1$) overestimates the converged frequency by only 3.3 %, and the 4-DOF model ($N = 2$) is within 0.1 %. Third, the canonical five-mode set ($n = 2, \ldots, 6$) used for the identifiability analysis lies well above the convergence threshold. For all practical purposes, the single-term Ritz basis is sufficient for the spectral and inverse conclusions drawn in this paper.

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Table 1: Cross-application evidence for the geometry–spectrum–inference thesis. Here f_2 denotes the lowest flexural mode, R_{eq} the equivalent radius where used, and κ the condition number of the scaled Jacobian for the local inverse problem.

Application	Geometry and key parameters	Characteristic frequencies	Coupling / inference takeaway
Abdomen	Canonical oblate shell: $a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 0.01\,\mathrm{m}$, $E = 0.1\,\mathrm{MPa}$, $\nu = 0.45$, $\rho_f = 1020\,\mathrm{kg/m^3}$, $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$	$f_2 = 3.95\,\mathrm{Hz}$; higher flexural modes in the 4 Hz to 10 Hz band; breathing mode $n = 0$ near 2490 Hz	Geometry filters forcing as well as spectra: the reported mechanical-to-airborne displacement ratio is $R \approx 3.3 \times 10^4$ in the canonical comparison; the same geometry underpins the identifiability study with $\kappa_{\mathrm{oblate}} = 69.4$
Bladder	Fill-dependent near-spherical shell over 50 mL to 500 mL; representative 300 mL state: $R = 4.15\,\mathrm{cm}$, $h = 1.79\,\mathrm{mm}$, $E = 70.9\,\mathrm{kPa}$	U-shaped $f_2(V)$ curve with a minimum $f_2 = 13.5\,\mathrm{Hz}$ at 222 mL; representative 300 mL value $f_2 = 13.9\,\mathrm{Hz}$	Geometry and strain-stiffening jointly organise the spectrum; mechanical coupling exceeds airborne coupling by $\mathcal{R}_{\mathrm{bladder}} \approx 6{,}400$; useful fill-state inference is possible, but no full geometry–material inversion is claimed
Watermelon	Oblate fruit approximated by an equivalent sphere: $a = 0.158\,\mathrm{m}$, $c = 0.123\,\mathrm{m}$, $h = 15\,\mathrm{mm}$, $R_{\mathrm{eq}} = 0.1453\,\mathrm{m}$, $E_{\mathrm{rind}} = 10\,\mathrm{MPa}$ to $250\,\mathrm{MPa}$	Tap-tone flexural frequency $f_2 = 46.4\,\mathrm{Hz}$ to $203.2\,\mathrm{Hz}$ across the constructed stiffness sweep; canonical example $f_2 = 89.9\,\mathrm{Hz}$ at $E_{\mathrm{rind}} = 50\,\mathrm{MPa}$	Equivalent-sphere reduction is forward-useful when geometry is measured independently and the inverse target is one effective stiffness parameter; that same reduction is unsuitable for joint recovery of (a, c, E)
Borborygmi	Quasi-spherical or slug-like gas pockets in a 30 mm lumen; canonical constrained-bubble parameters: $V = 1\,\mathrm{mL}$ to $50\,\mathrm{mL}$, $E = 10\,\mathrm{kPa}$, $h = 3\,\mathrm{mm}$	Constrained-bubble resonance 135 Hz to 440 Hz; canonical 10 mL pocket: 223 Hz; healthy spectral peak near 340 Hz implies $V \sim 3\,\mathrm{mL}$	Geometry selects the active resonance family; narrow inference about pocket size or mechanism is plausible, but only because the inverse task is low-dimensional rather than a full shell identification problem

Table 2: Convergence of the $n = 2$ flexural frequency with basis enrichment. N is the number of Legendre terms per displacement component; Δf is the relative deviation from the $N = 8$ reference value.

N (DOFs = $2N$)	f_2 (Hz)	Δf vs. N=8
1 (2)	3.800	+3.3%
2 (4)	3.682	+0.1%
3 (6)	3.679	$< 0.01\%$
≥ 4 (≥ 8)	3.678	converged